

Beans



Bean means an object

- Spring creates automatically
- Spring configures based on our needs
- Spring controls and manages its entire lifecycle
- Instead of creating object again and again, Spring creates once and reuses it.

Beans are the main building blocks inside a Spring application.

Spring connects these beans together to form the complete application.

How to create Beans?

1. Stereotype annotations

- @Component → Marks class as bean
- @Service → Bean for service layer
- @Repository → Bean for DB layer
- @Controller → Bean for web controllers
- @Bean → Manually create bean in config class

2. Using @Bean in a @Configuration Class

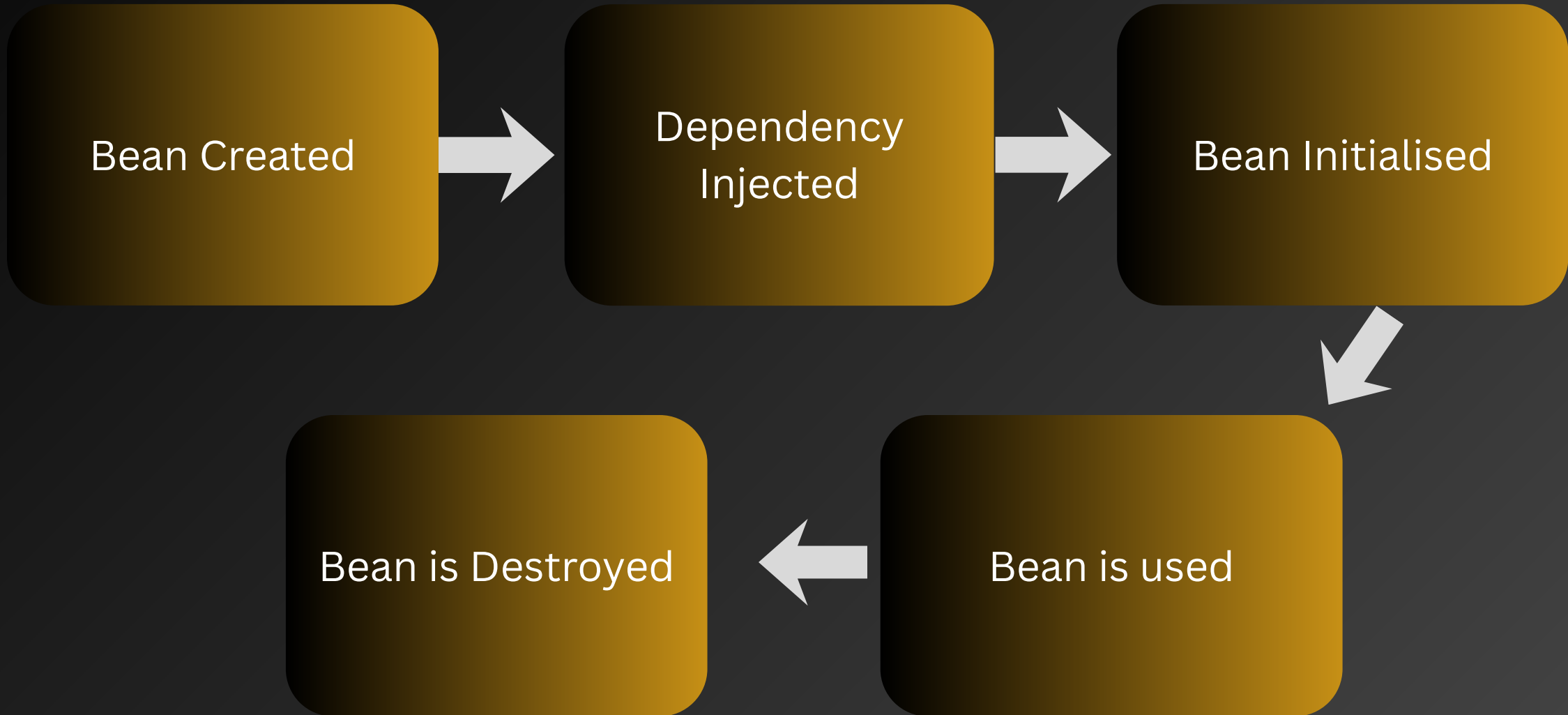
IOC Container and Application Context

IOC Container - The Bean Factory

Role of IoC Container in Spring:

- The IoC Container is the heart of the Spring Framework
- It creates and manages all objects in the application — these objects are called Beans.
- It injects dependencies automatically, so we don't need to create objects manually using new.
- It connects and configures beans with each other.
- It controls the entire lifecycle of beans — from creation to destruction.
- In simple words: IoC Container handles object creation, dependency injection, and lifecycle management for the developer.

Lifecycle of a Bean



Bean Lifecycle Methods

@PostConstruct

- This annotation is applied to a method that should run right after the bean is created.
- The method executes after Spring injects all required dependencies.
- It is typically used to initialize resources or setup configuration before the bean starts working.

@PreDestroy

- This annotation marks a method that runs just before the bean is removed from the IoC container.
- Used to close connections, release resources, or perform cleanup tasks.
- Ensures that everything is properly shutdown before the application stops.

Bean Scopes

Scope	Description (One Line)
Singleton	Only one instance per Spring container (default scope).
Prototype	Creates a new instance every time it is requested.
Request	Creates one bean per HTTP request (Web applications only).
Session	Creates one bean per user session.
Application	One bean per servlet context for the entire app.
WebSocket	One bean per WebSocket session.